

Mutasim Billah

ICT Undergraduate · Robotics & Embedded Systems

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SUMMARY

ICT undergraduate at Bangladesh University of Professionals who designs and builds robots and ESP32-based devices, from wiring and firmware to competition day. Robo Soccer runners-up at the National Robotics Championship 2025. Learning PCB design and full-stack web development; seeking internships in embedded systems, robotics, or software.

EDUCATION

B.Sc. in Information and Communication Engineering
Bangladesh University of Professionals (BUP), Dhaka

2023 – Present

TECHNICAL SKILLS

Languages	C, C++, Java, Python, SQL, HTML, CSS
Embedded & hardware	ESP32, Arduino, ESP-IDF, PD motor control, sensor arrays, I ² C/SPI displays, nRF24 radios
Software & tools	Git, VS Code, Flask, SQLAlchemy, MySQL, PostgreSQL, MATLAB, Cisco Packet Tracer
Learning	PCB design, React, TypeScript, Node.js

PROJECTS

CPU Scheduler Visualizer [GitHub](#)

C++ · ESP32

- Handheld device that teaches CPU scheduling on two displays (SSD1306 OLED, 240×240 ST7789 IPS) and four buttons.
- Five algorithms (FCFS, SJF, SRTF, Priority, Round Robin) with step-by-step playback, a live Gantt chart, waiting, turnaround, and response-time metrics, and a Compare All mode.

Robo Soccer Robot

Designer and builder

- Designed and built a soccer robot from scratch and tested it for competition: runners-up at NRC 2025, 4th at IUBAT.

Line-Following Robots (Mark-X, Head Hunter)

Hardware and firmware

- Made physical modifications, software changes, and firmware updates across several build iterations.
- Mark-X firmware: PD control over a 16-sensor array read through an analog multiplexer, with a state machine for gaps, junction priority, inverted-line sections, and a finish-box auto-stop.

ESP32 Wireless Signal Analyzer and Smart Clock

ESP32

- Wireless signal analyzer with an ST7789 display for Wi-Fi channel analysis and Bluetooth scanning.
- Smart clock on an SSD1306 OLED with weather integration, animations, and power-saving features.

Web Applications

Python · Flask · Java

- Art Attack: Flask and SQLAlchemy art marketplace with team battles, moderation, checkout, and PDF receipts.
- Restaurant Management System: Java web app with staff roles, reservations, orders, and billing.
- Q-Less Printing (in progress): printing service for BUP built with React, TypeScript, Node.js, and PostgreSQL.

LEADERSHIP AND ACTIVITIES

Technical Team Head

Circuit Clash 1.0

- Led technical operations for a national-level engineering competition: setup, troubleshooting, and live coordination.

Technical Lead

SafetyPod

- Technical lead for the team developing SafetyPod, a safety device.

Workshop Contributor

Soccer Bot Workshop

- Helped participants learn robot movement, control systems, and hardware integration in hands-on sessions.

ACHIEVEMENTS

- Runners-up, Robo Soccer, National Robotics Championship 2025.
- Top 10, Robo Fusion (UFTB) with Team DurontoJatra; qualified for the World SDG Challenge 2026, Kuala Lumpur.
- 4th place, Soccer Bot segment, IUBAT Robotics Competition · 5th place, Machine Mania 3.0, BUTEX.
- Exhibited AgroBot with Team QuadBits at Bangladesh Innovation Fair 2026.
- Silver and bronze medals in taekwondo at the National Games, 2016 and 2017.